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Exercises — Theme 1

Statements

Exercise 1 : Stern–Gerlach and the idea of quantization

Keywords: Stern–Gerlach, spin, measurement postulate, experiment

Question. A beam of atoms crosses a Stern–Gerlach device along z ; the “spin-up” component is kept. That sub-beam enters a second device, also along z . What is seen on the screen?

Question. Same preparation, but the second device is along x . What happens?

Hints

Exercise 1 : Stern–Gerlach and the idea of quantization

Hint

Think about repeating a projective measurement in the same basis.

Solutions

Exercise 1 : Stern–Gerlach and the idea of quantization

Solution

The atoms were already projected onto $|+_z\rangle$; the second filter does not split that component further: a single spot, consistent with the preparation.

Solution

$|+_z\rangle$ expands in the x basis with two nonzero components; two spots appear, each with probability $1/2$ (barring degenerate magnet alignment).